

- 6 (a) Define electric potential at a point.

.....

.....

..... [2]

- (b) An isolated conducting sphere of radius  $R$  is in a vacuum. The sphere has a charge of  $+Q$  distributed around its surface, as illustrated in Figure 6.1.

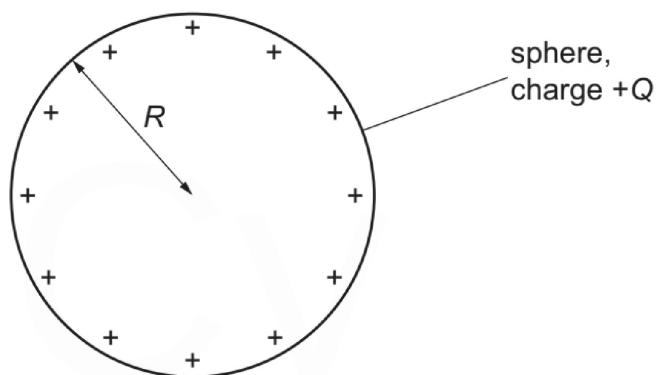


Figure 6.1

- (i) State an expression, in terms of  $R$  and  $Q$ , for the magnitude  $V_0$  of the electric potential at the surface of the sphere. Identify any other symbols that you use.

$V_0 =$  ..... [1]

- (ii) On Figure 6.2, sketch the variation of the electric potential  $V$  at distance  $x$  from the centre of the sphere between  $x = R$  and  $x = 3R$ .

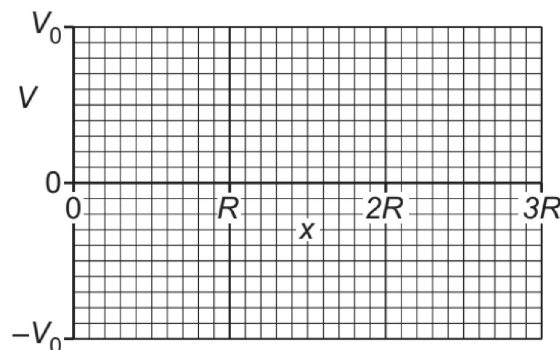


Figure 6.2

- (c) The dome of an electrostatic generator, called a Van de Graaff generator, may be considered to be a conducting sphere of radius 11.9 cm.

The dome is charged so that the electric potential at its surface is  $-4.80 \times 10^4 \text{ V}$ .

- (i) Calculate the charge on the surface of the dome.

charge = ..... C [2]

- (ii) Explain why the dome acts as a capacitor.

.....

..... [1]

- (iii) Calculate the capacitance of the dome. Give a unit with your answer.

capacitance = ..... unit ..... [2]

- (iv) Determine the energy stored by the dome.

energy = ..... J [2]

[Total: 13]